

SystemC Debugging in DVE

Example Instructions

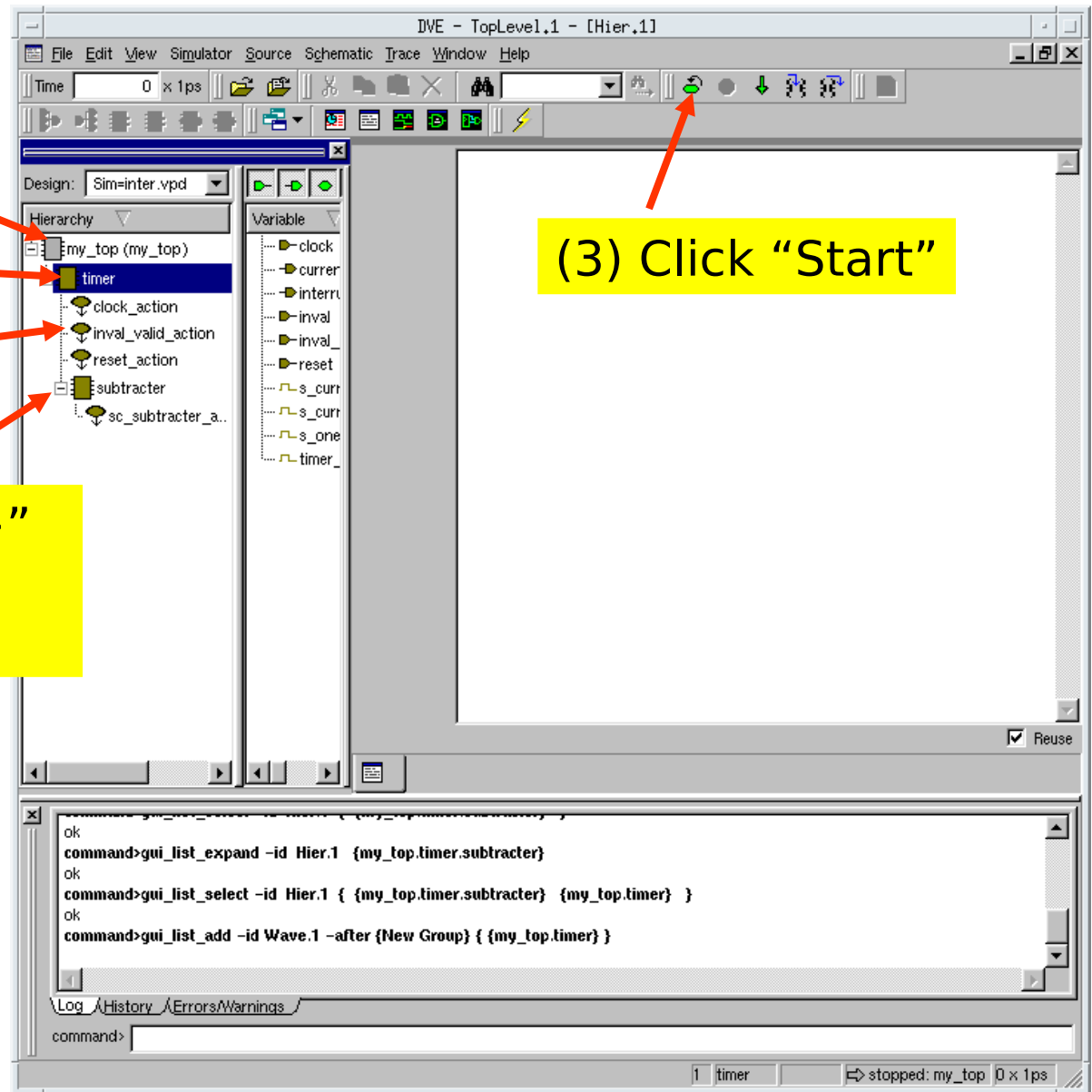
Verilog instance

SystemC
instance
SystemC
process

(4) Click on “+”
to open
hierarchy

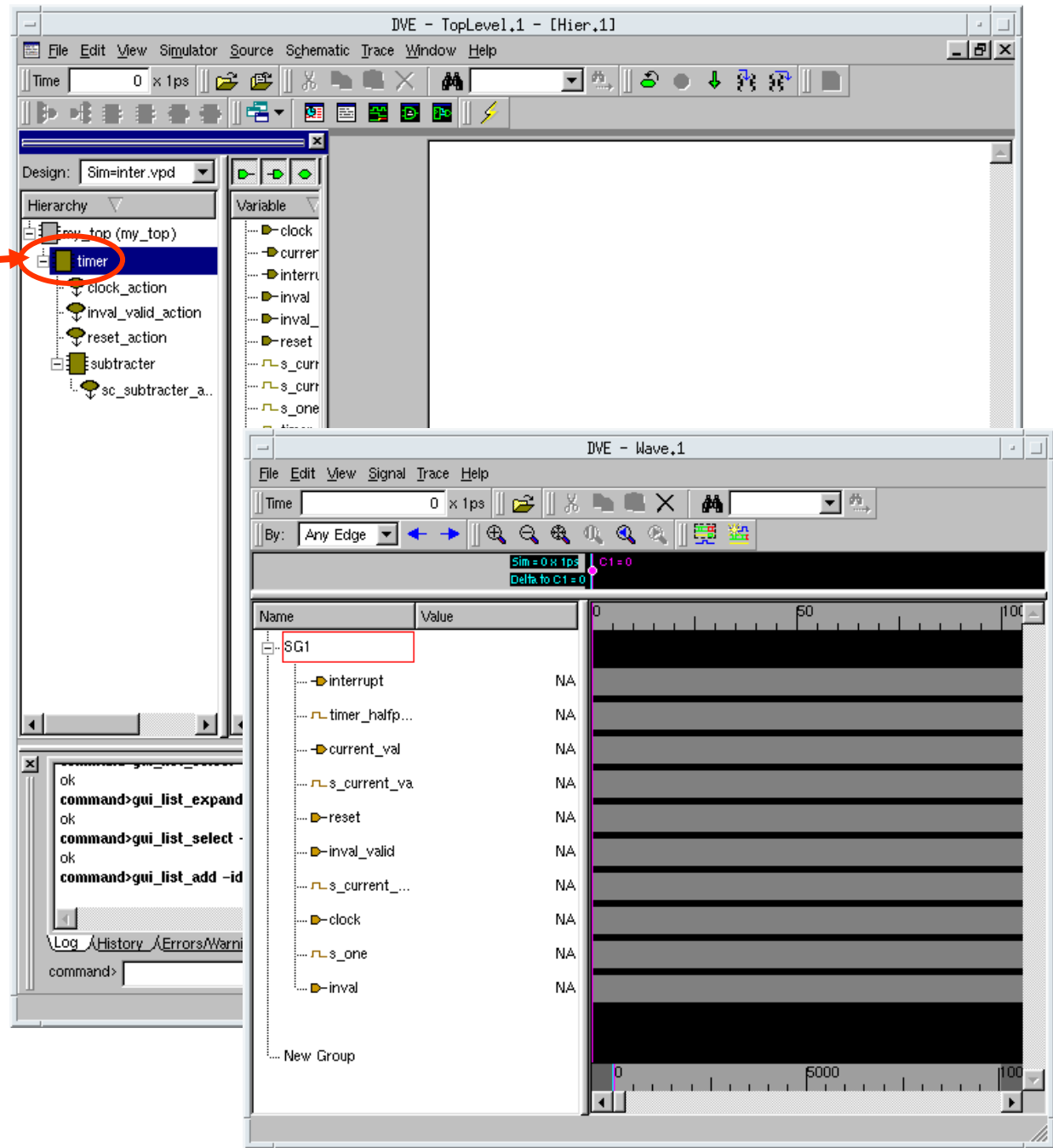
(1) compile:
shell> comp.csh

(2) Start DVE:
shell> simv -gui



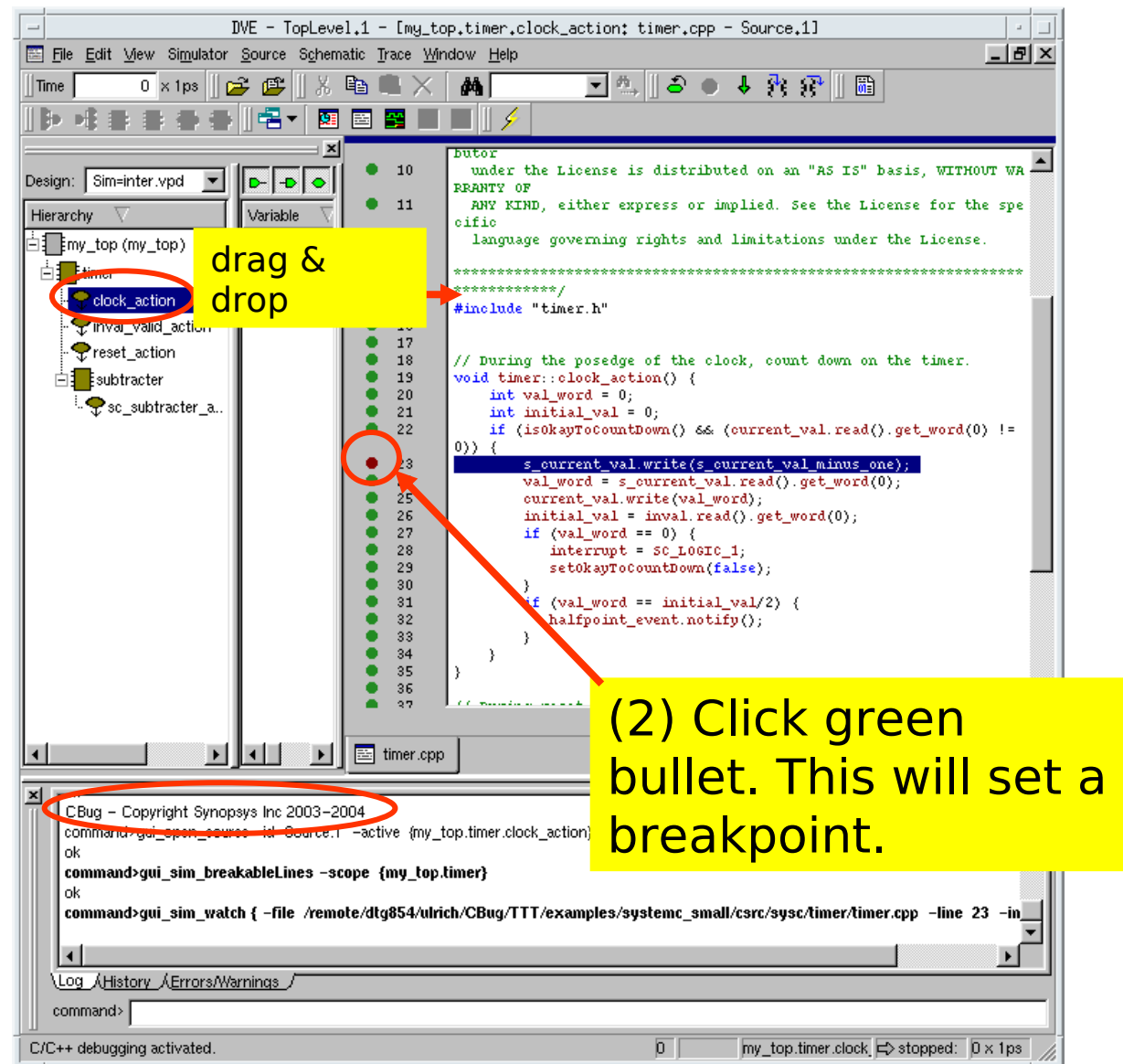
(1) Right-mouse-click “timer”, select “Add to Waves”

→ Waveform window appears
(2) Ditto, add “my_top” to waves



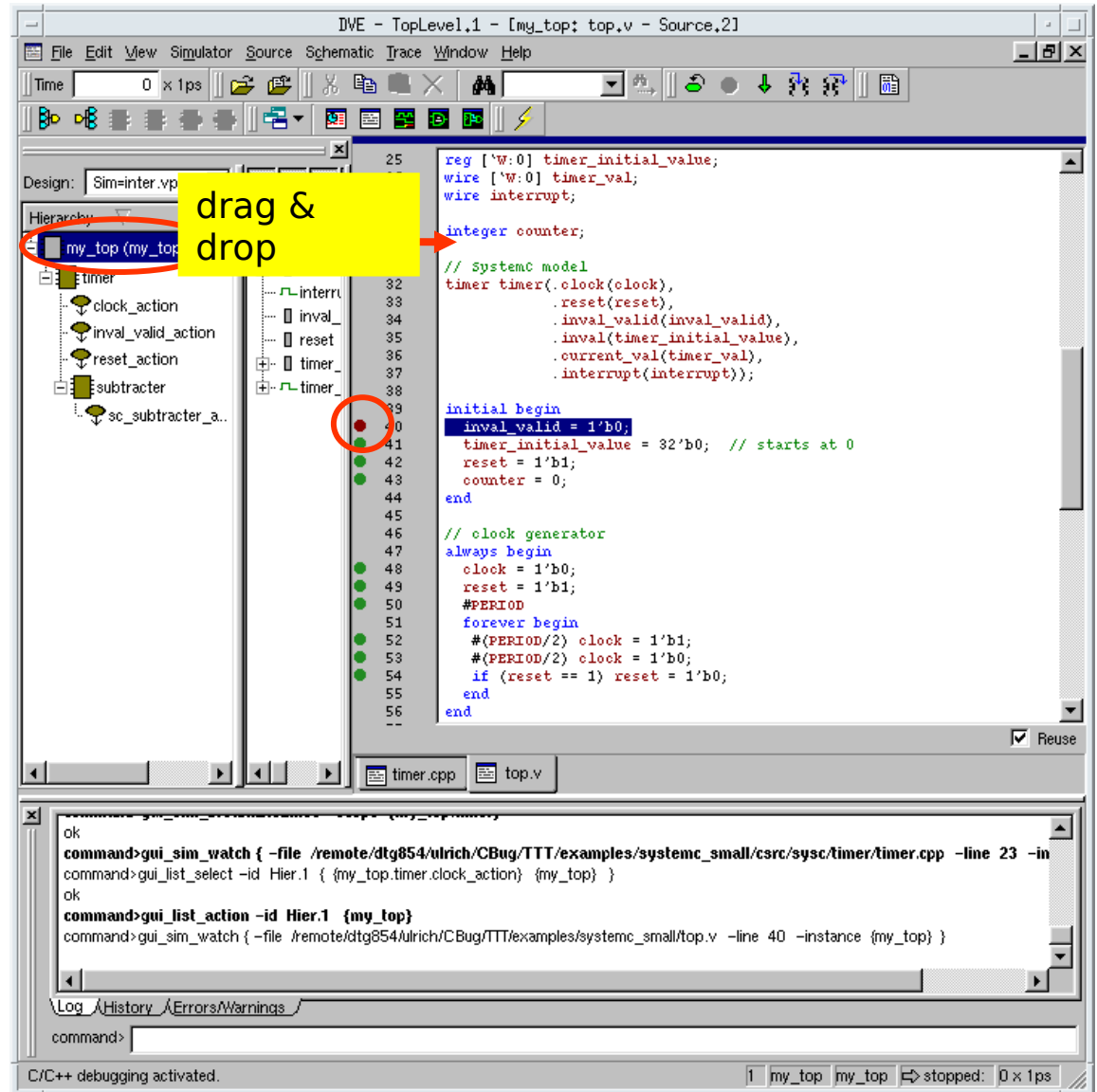
(1) Select
“clock_action”
SystemC
process and
drag into source
window

➔ CBug starts
automatically.
Wait until done
and SystemC
source code
appears (it will
take a few
seconds)



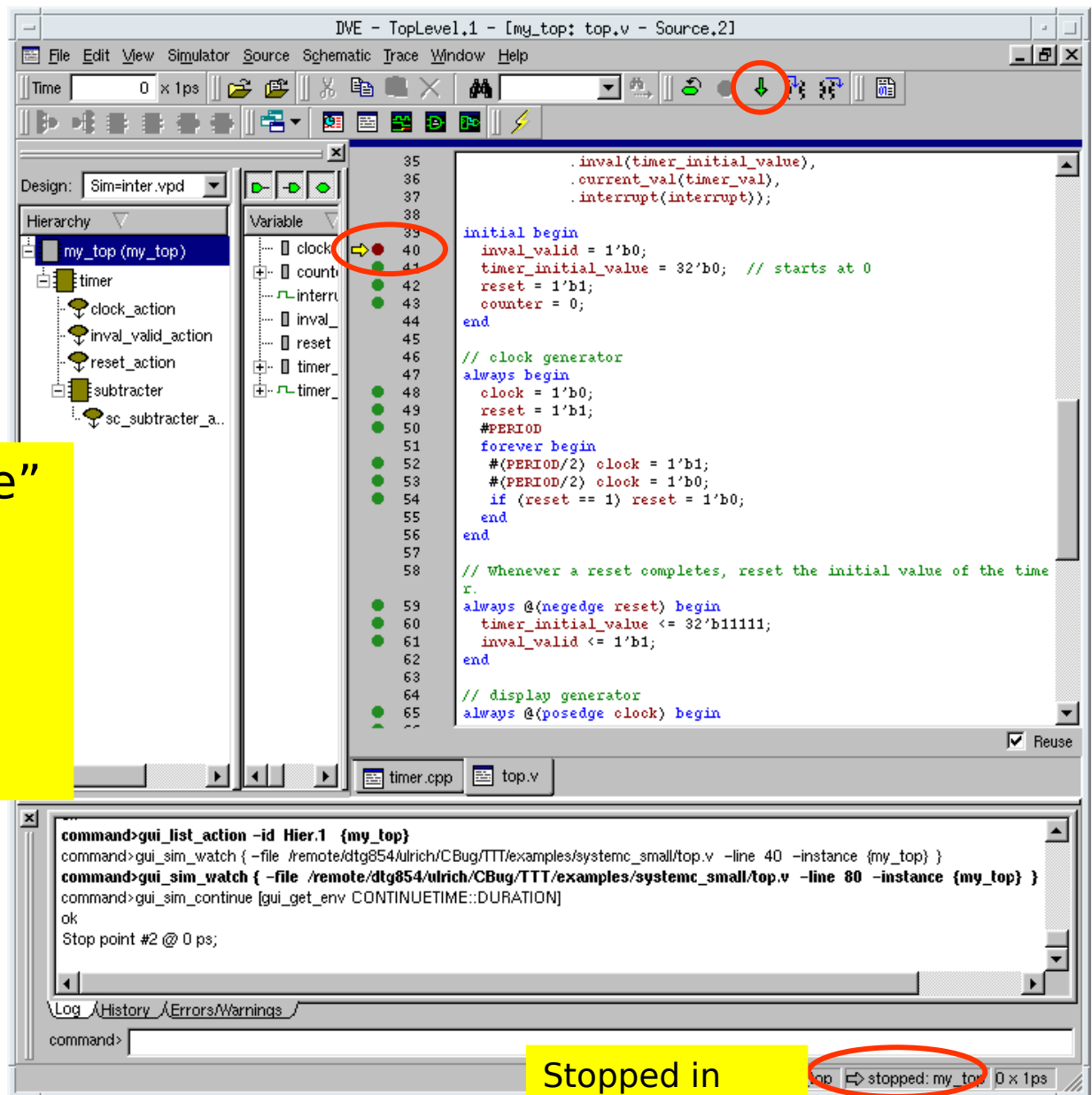
(1) Open top-level Verilog instance, set breakpoint

(2) Set breakpoints in lines 40 and 80



(1) Click "Continue" button

→ Simulation will stop in Verilog first



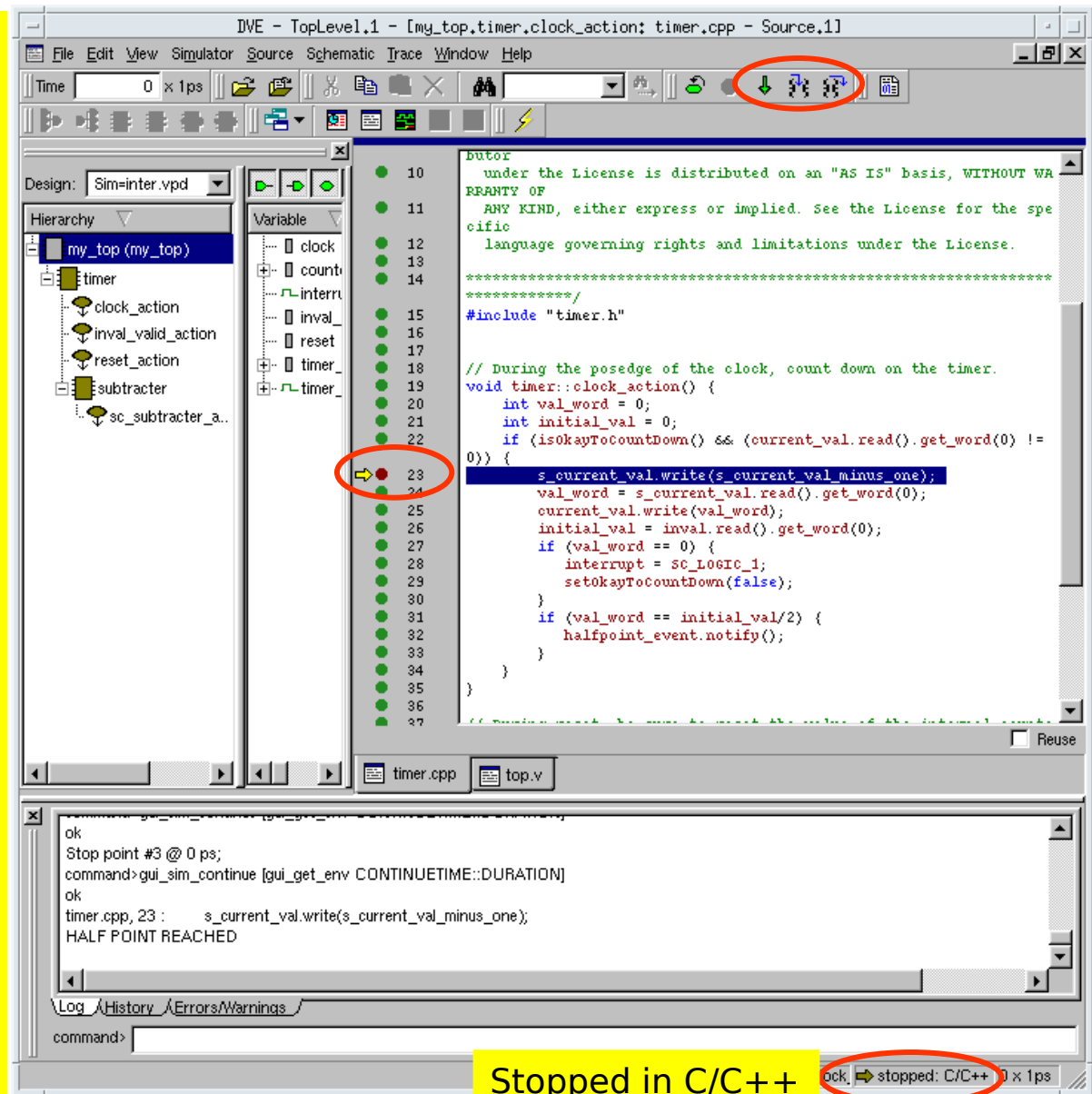
(1) Click “Continue” button again

→ Simulation will stop in SystemC model

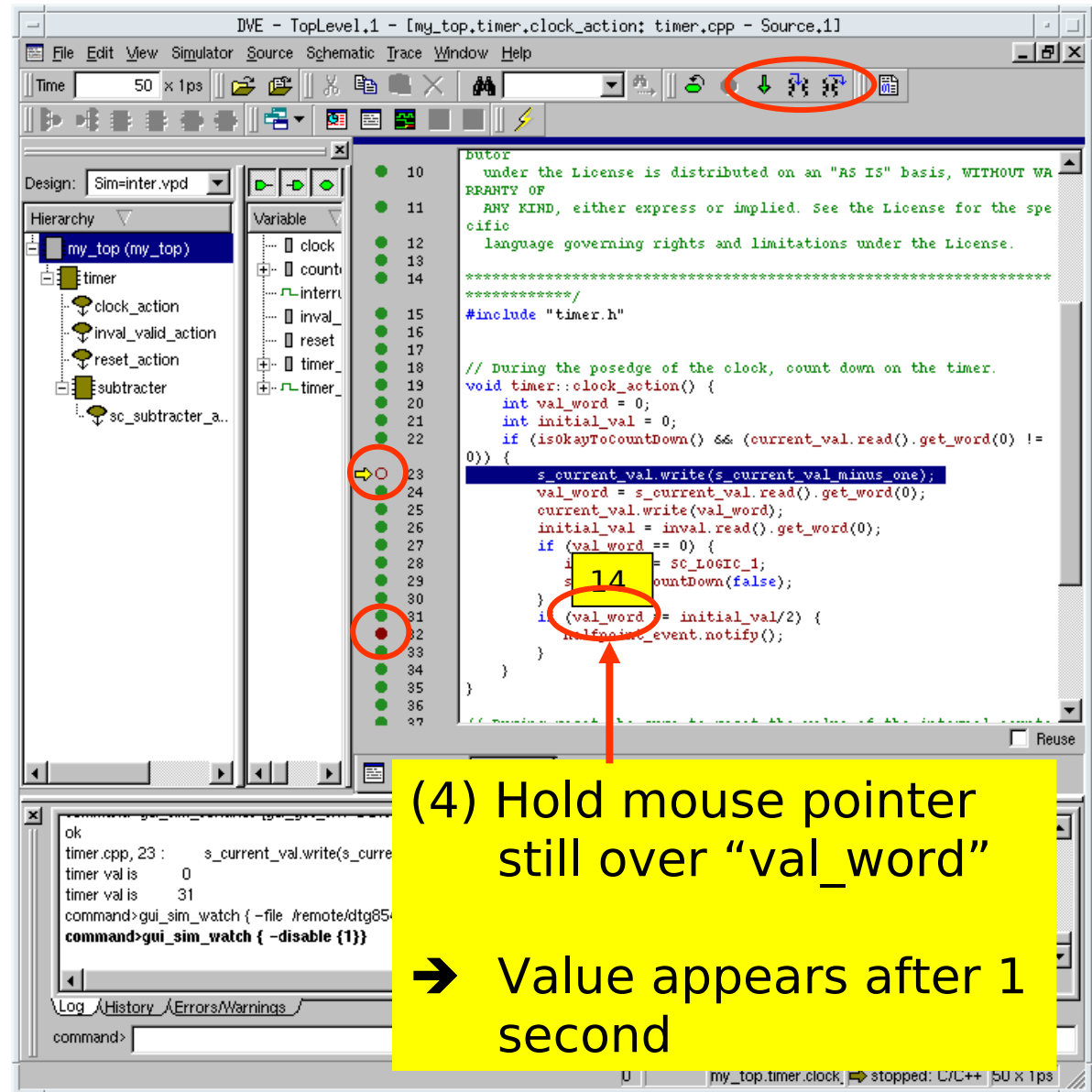
(2) Click “Step” or “Next” buttons.

→ “Next” executes one source line

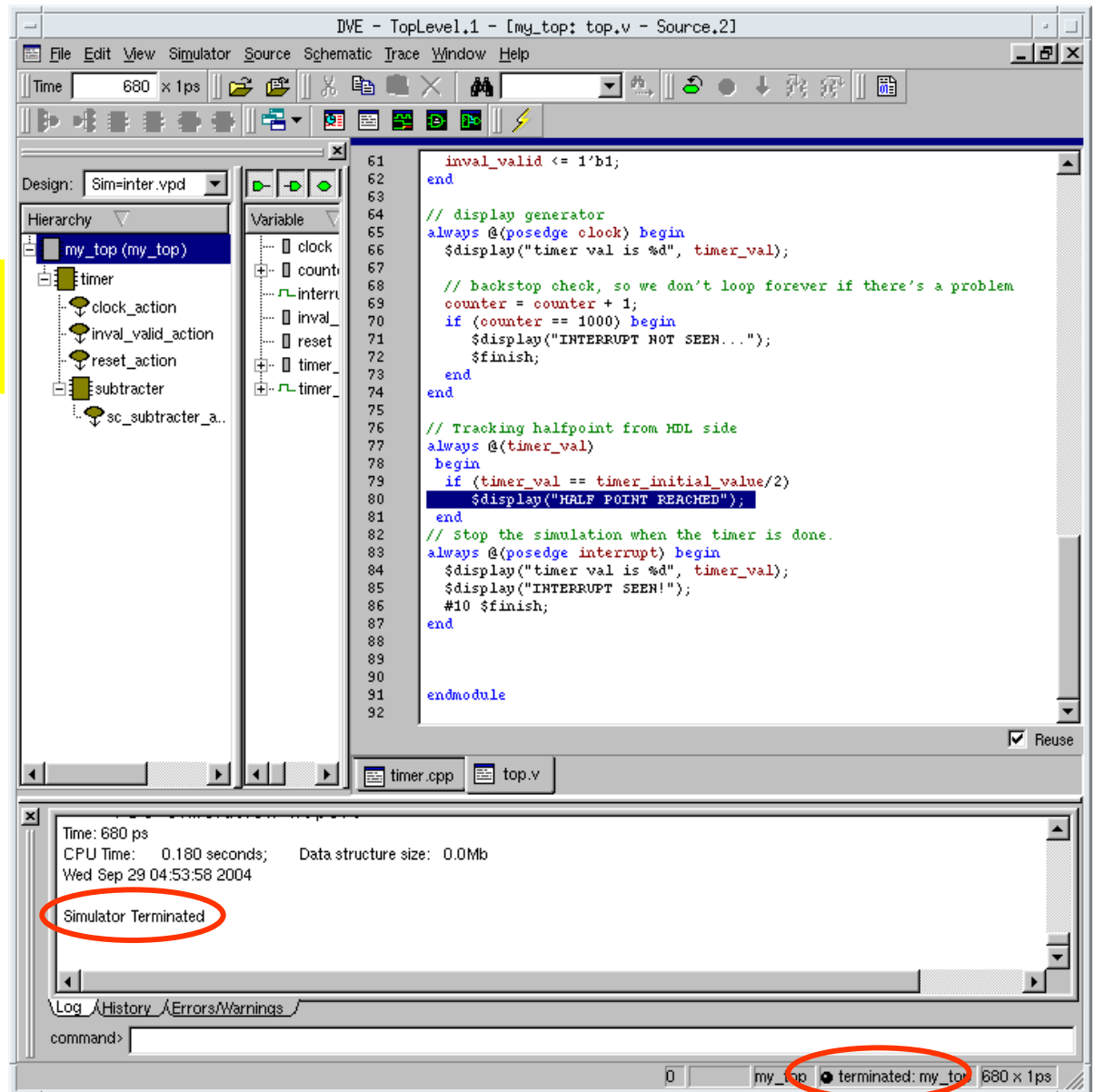
→ “Step” does the same but will jump into functions (if any)



- (1) Disable breakpoint in line 23 by clicking on bullet
- (2) Set another breakpoint in line 32
- (3) Continue to line 32



(1) Continue until simulation ends



(1) Check waveform

